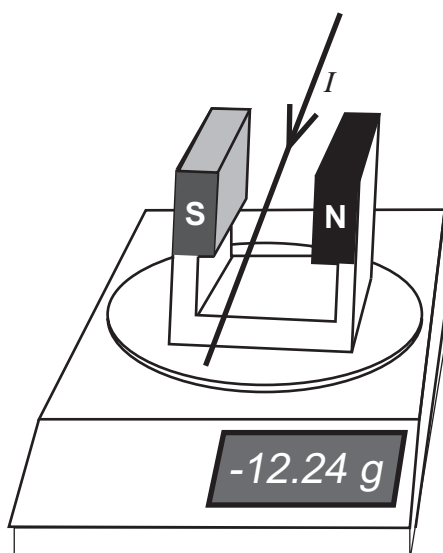


4. A student carries out an investigation. She uses a top pan balance to investigate the relationship $F = BIl$ using the set up shown in the diagram. The current-carrying wire is placed perpendicular to the magnetic field, B .

The current in the wire is varied using a variable power supply while the magnetic field, B , and the length of wire in the field, l , are kept constant. Before switching on the current, the balance is set to zero.



- (a) (i) Explain why the reading on the balance is negative for the direction of current and magnetic field shown in the diagram. [3]

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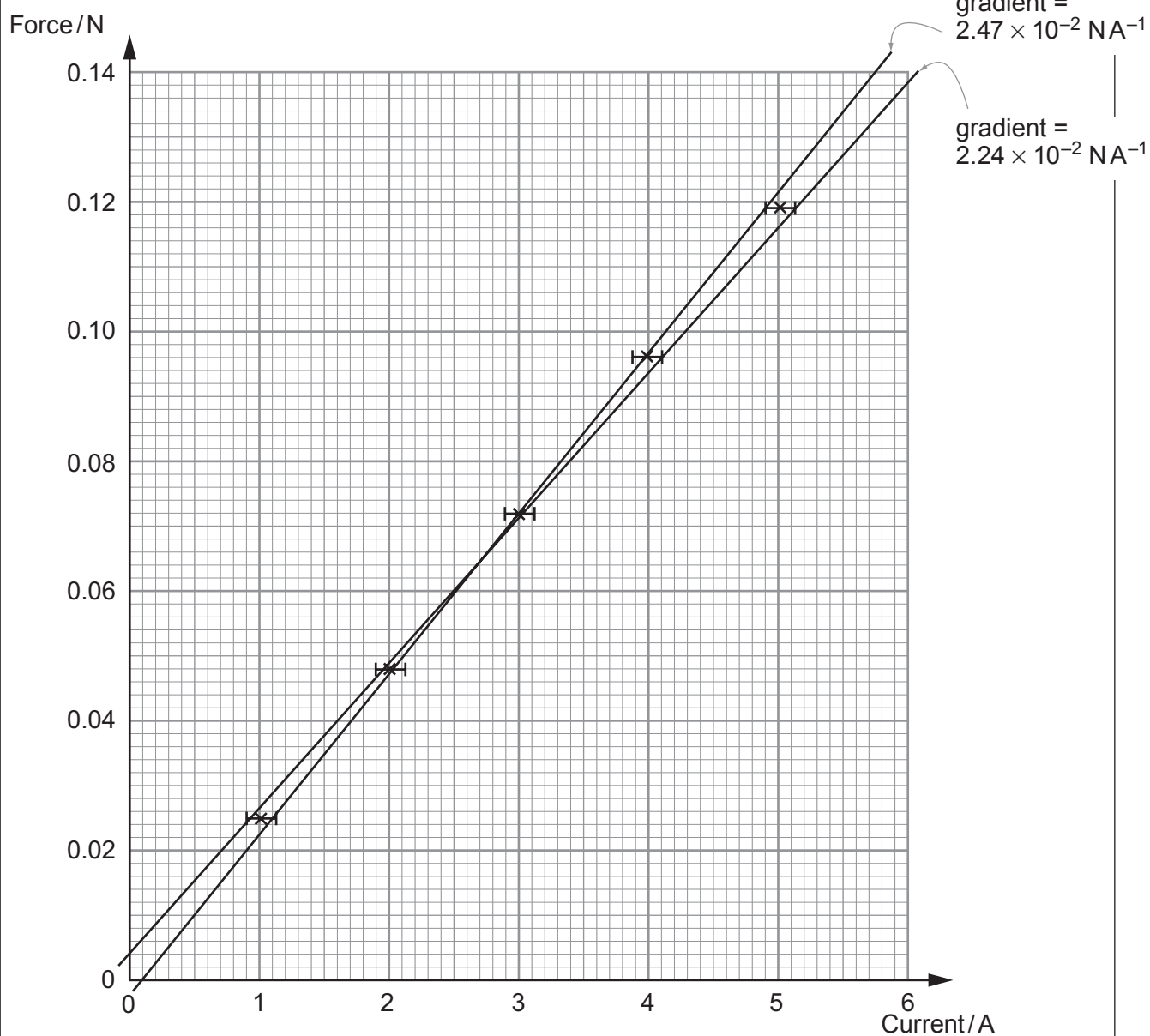
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The student's results are plotted on the grid.



- (ii) Calculate the mean value of the gradients ($2.47 \times 10^{-2} \text{ NA}^{-1}$ and $2.24 \times 10^{-2} \text{ NA}^{-1}$) and its **percentage** uncertainty. [3]

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- (iii) The length of wire in the uniform magnetic field is (8.0 ± 0.5) cm. Determine a value for the magnetic field strength and its **absolute** uncertainty, stating both to an appropriate number of significant figures. [4]

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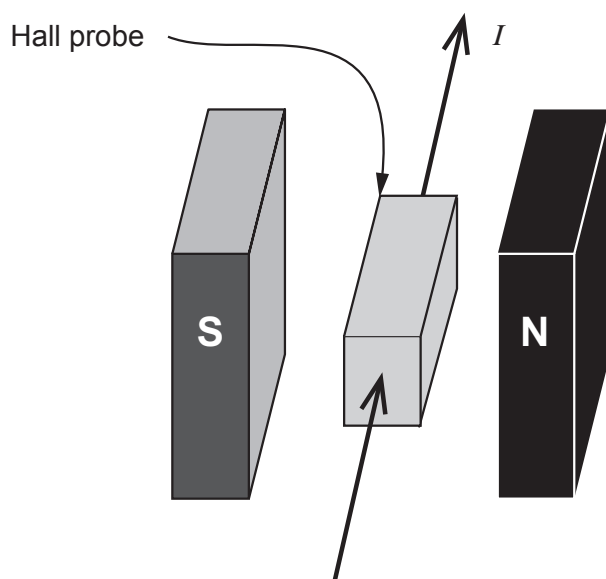
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- (b) In another experiment a Hall probe is used to confirm the value of the magnetic field strength obtained. The value of B obtained using the Hall probe is (310 ± 8) mT.



- (i) Explain how the Hall voltage arises in the Hall probe shown. [3]

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- (ii) Discuss the effectiveness of the experiment in part (a) for confirming the relationship $F = BIl$ and measuring the magnetic field strength, compared with using the Hall probe {see (a)(iii) and (b)}. [6 QER]

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